

❏ 1. Patches in Linux

❏ What is a Patch?

- A **patch** is a small update to fix bugs, add features, or address security issues in software or kernel.
- Can be applied to **source code**, **packages**, or **binaries**.

❏ Types of Patches:

- **Security patches:** Address vulnerabilities (e.g., CVEs)
- **Bug fixes:** Correct software errors
- **Feature patches:** Add/modify functionality
- **Kernel patches:** Modify Linux kernel (e.g., kpatch, kgraft)

❏ Applying Patches to Source Code:

```
bash
```

Copy code

```
patch < original.patch
```

❏ Create Patch:

```
bash
```

Copy code

```
diff -u original_file updated_file > patch.diff
```

❏ Patch Management Tools:

Tool	Function
patch	Applies patch files to source code
yum update / apt upgrade	Apply package updates (with patches)
kpatch, kgraft	Live kernel patching (no reboot needed)
dnf-automatic, unattended-upgrades	Automate patch application

❏ 2. System Management

System management in Linux involves administering system-wide configurations, processes, users, updates, and services.

❏ System Management Tasks:

1.

User & Group Management

- useradd, passwd, usermod, groupadd

2.

Service Management

- systemctl, service

3.

Package Management

- apt, dnf, yum, zypper, pacman

4.

Scheduled Jobs

- cron, at, systemd timers

5.

Disk and File System Management

- lsblk, mount, fdisk, lvm, df, du

6.

Monitoring Tools

- top, htop, vmstat, iostat, free, uptime

7.

System Logs

- Stored in /var/log/ (e.g., syslog, auth.log, dmesg, messages)

8.

Backup and Recovery

- rsync, tar, timeshift, Deja Dup

9.

Networking

- ip, nmcli, hostname, ifconfig, netstat

10.

Boot and Kernel Management

- grub, initrd, systemd, sysctl

☒ 3. X Configuration Server (X.Org / X11)

☒ What is X Server?

- X server (commonly **X.Org Server**) is the core component of the Linux graphical system.
- It handles:
 - Input devices (mouse, keyboard)
 - Output (display)
 - Window drawing and movement

☒ Key Configuration Files:

File	Description
/etc/X11/xorg.conf	Manual X server config (rarely used now)
/etc/X11/xorg.conf.d/	Additional config snippets

~/.xinitrc	User's X session setup
~/.Xresources	GUI appearance settings

☒ **Commands:**

bash

Copy code

startx # Start X session xrandr # Configure resolution, screens xinput # Manage input devices glxinfo
Info about OpenGL support xev # Event tester

☒ **Display Managers:**

- GUI login managers (e.g., GDM, LightDM, SDDM)

☒ Desktop Environments:

- Built on X (e.g., GNOME, KDE, XFCE)

☒ Wayland:

- Modern alternative to X11 with better security and performance
- Many distros now support both (with Wayland as default in GNOME)